

Numeric

Numeric data types:-

1. Integer
2. float
3. complex

Integer data type:

The int data type represents values or numbers without decimal values. In python, there is no limit for the int data type. It can store very large values conveniently.

a=10

type(a) O/P:- `<class 'int'>`

Note: In Python 2nd version long data type was existing but in python 3rd version long data type was removed.

We can represent int values in the following ways

1. Decimal form (by default)

2. Binary form

3. Octal form

4. Hexa decimal form

1. Decimal form(base-10):

It is the default number system in Python. The allowed digits are: 0 to 9

Ex: a =10

2. Binary form(Base-2):

The allowed digits are : 0 & 1

Literal value should be prefixed with 0b or 0B

Eg: a = 0B1111

a =0B123

a=b111

3. Octal Form(Base-8):

The allowed digits are : 0 to 7

Literal value should be prefixed with 0o or 0O.

Ex: a=0o123

a=0o786

4. Hexa Decimal Form(Base-16):

The allowed digits are : 0 to 9, a-f (both lower and upper cases are allowed)

Literal value should be prefixed with 0x or 0X.

Ex: a=0X9FcE

a=0x9aDF

Note: Being a programmer we can specify literal values in decimal, binary, octal and hexa decimal forms. But PVM will always provide values only in decimal form.

```
# binary data type(Base-2)
x= 0b1111
y= 0B1010

# by default converted into decimal
print(x) # O/P-15
print(y) # O/P-10

# octal data type(Base-8)
x=0o765
y=00542

# by default converted into decimal
print(x) # O/P-501
print(y) # O/P-354

# decimal data type(Base-10)---default
x=10
y=50

# by default converted into decimal
print(x) # O/P-10
print(y) # O/P-50

# Hexadecimal data type(Base-16)
x=0X8EA
z=0x5eA

# by default converted into decimal
print(x) # O/P-2282
print(y) # O/P-50
```

Base Conversions: Python provide the following in-built functions for base conversions

```
# Base Conversions
# bin()
print(bin(15)) # o/p- 0b1111
print(bin(0o11)) # o/p- 0b1001
print(bin(0X10)) # o/p- 0b10000
```

```
# oct()
print(oct(10)) # o/p-0o12
print(oct(0B1111)) # o/p-0o17
print(oct(0X123)) # o/p-0o443

# hex()
print(hex(100)) # o/p-0x64
print(hex(0B111111)) # o/p-0x3f
print(hex(0o12345)) # o/p- 0x14e5
```

Float Data Type in Python:

The float data type represents a number with decimal values. floating-point numbers can also be written in scientific notation. e and E represent exponentiation. where e and E represent the power of 10. For example, the number $2 * 10^{pow2}$ is written as 2E2, such numbers are also treated as floating-point numbers.

```
salary = 50.5
print(salary)
print(type(salary))

O/P---
50.5
<class 'float'>
```

Example: Print float values

```
a = 2e2 # 2*10^2  e stands for 10 to the power
b = 2E2 # 2*10^2
c = 2e3 # 2*10^3
d = 2e1
print(a)
print(b)
print(c)
print(d)
print(type(a))

O/P----
200.0
200.0
2000.0
20.0
<class 'float'>
```

Complex Data Type in python:

The complex data type represents the numbers that are written in the form of $a+bj$ or $a-bj$, here a is representing a real part of the number and b is representing an imaginary part of the number. The suffix small j or upper J after b indicates the square root of -1 . The part “ a ” and “ b ” may contain integers or floats.

```
a = 3+5j
b = 2-5.5j
c = 3+10.5j
print(a)
print(b)
print(c)
print()
print("A+B=", a+b)
print("B+C=", b+c)
print("C+A=", c+a)
print("A*B=", a*b)
print("B*C=", b*c)
print("C*A=", c*a)
print("A+B+C=", a+b+c)
print("A/B=", a/b)
```

O/P---

```
(3+5j)
(2-5.5j)
(3+10.5j)
```

```
A+B= (5-0.5j)
B+C= (5+5j)
C+A= (6+15.5j)
A*B= (33.5-6.5j)
B*C= (63.75+4.5j)
C*A= (-43.5+46.5j)
A+B+C= (8+10j)
A/B= (-0.6277372262773723+0.
```